

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Wednesday 18 June 2025

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/3A**

Further Mathematics

Advanced

PAPER 3A: Further Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P76379A

©2025 Pearson Education Ltd.
Y:1/1/1/1/



1. (a) Given that

$$y = e^{\operatorname{cosec}^2 x}$$

complete the table below with the value of y corresponding to $x = 2$, giving your answer to 2 decimal places.

x	1.5	1.75	2	2.25	2.5
y	2.73	2.81		5.22	16.31

(1)

- (b) Use Simpson's rule, with all the values of y in the completed table, to estimate, to one decimal place, the value of

$$\int_{1.5}^{2.5} e^{\operatorname{cosec}^2 x} dx$$

(3)

- (c) Using your answer to part (b) and making your method clear, estimate the value of

$$\int_{1.5}^{2.5} e^{\cot^2 x} dx$$

(2)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. $\left[\begin{array}{l} \text{The Taylor series expansion of } f(x) \text{ about } x = a \text{ is given by} \\ f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \dots + \frac{(x - a)^r}{r!}f^{(r)}(a) + \dots \end{array} \right]$

Given that

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 2xy = 4 \quad (\text{I})$$

(a) show that

$$\frac{d^4y}{dx^4} = a\frac{dy}{dx} + bx\frac{d^2y}{dx^2} + c\frac{d^3y}{dx^3}$$

where a , b and c are integers to be determined.

(4)

Hence, given that $y = 1$ and $\frac{dy}{dx} = 1$ when $x = 2$

- (b) determine a Taylor series solution, in ascending powers of $(x - 2)$, up to and including the term in $(x - 2)^4$, of the differential equation (I), giving each coefficient in simplest form.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)



3. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Use algebra to determine the values of x for which

$$\frac{x^2 - 4}{|x - 5|} > 4x + 8$$

(6)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4.

$$y = \sqrt{(15 + e^{2x})}$$

(a) Show that

$$\frac{d^2y}{dx^2} = \frac{e^{2x}(e^{2x} + 30)}{(15 + e^{2x})^{\frac{3}{2}}} \quad (3)$$

(b) Hence determine the Maclaurin series expansion for y , in ascending powers of x , up to and including the term in x^2 , giving each term in simplest form. (2)

(c) Use the series expansion for $\sin x$ in ascending powers of x to determine, in simplest form, the first 2 non-zero terms in ascending powers of x of the series for $\sin 3x$. (1)

(d) Use the answers to parts (b) and (c) to show that

$$\lim_{x \rightarrow 0} \frac{\sqrt{(15 + e^{2x})} - 4}{\sin 3x} = \frac{1}{12} \quad (2)$$



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5. (a) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that

$$\int \frac{1}{1 + \operatorname{cosec} x} dx = \int \frac{4t}{(1 + t^2)(1 + t)^2} dt \quad (3)$$

- (b) Hence show that

$$\int \frac{1}{1 + \operatorname{cosec} x} dx = Ax + \frac{B}{1 + \tan\left(\frac{x}{2}\right)} + k$$

where A and B are constants to be determined and k is an arbitrary constant.

(5)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 8 marks)



6. (i) Use L'Hospital's Rule to show that

$$\lim_{x \rightarrow 0} \frac{1 - \cos 7x}{x \sin 9x} = \frac{49}{18} \quad (4)$$

(ii) $y = x^2 e^{3x}$

(a) Use Leibnitz's theorem to show that, for $k \in \mathbb{N}$

$$\frac{d^k y}{dx^k} = 3^{k-2} e^{3x} (Ax^2 + Bkx + Ck(k-1))$$

where A , B and C are integers to be determined.

(5)

(b) Hence determine the values of x for which

$$\frac{d^9 y}{dx^9} = 0 \quad (2)$$



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 11 marks)



7. The concentration, $P \text{ mg m}^{-3}$, of a pollutant in a reservoir, t days after the pollutant entered the reservoir, is modelled by the differential equation

$$\frac{1}{P} \frac{dP}{dt} = 4tP^2 - 1 \quad (\text{I})$$

- (a) Show that the transformation $x = \frac{1}{p^2}$ transforms equation (I) into the equation

$$\frac{dx}{dt} - 2x = -8t \quad (\text{II})$$

Given that $P = 0.5$ when $t = 0$

- (b) solve differential equation (II) to show that, according to the model,

$$P^2 = \frac{1}{4t + 2 + ke^{2t}}$$

where k is a constant to be determined. (6)

Given that the concentration of the pollutant in the reservoir, 3 days after the pollutant entered the reservoir, was 0.034 mg m^{-3}

- (c) comment on the reliability of the model, giving a reason for your answer. (2)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 11 marks)



8. The ellipse E has equation

$$\frac{x^2}{64} + \frac{y^2}{36} = 1$$

The line l has equation $y = mx + c$ where m and c are constants.

(a) Show that the x coordinates of the points of intersection of E and l satisfy the equation

$$(16m^2 + 9)x^2 + 32cmx + 16c^2 + k = 0$$

where k is a constant to be determined.

(2)

Hence, given that the line l

- is a tangent to E
- passes through the point $(10, 20)$

(b) determine the possible equations of l

(5)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 7 marks)



9. The parallelogram P has vertices E, F, G and H , with coordinates

$$E(10, -1, -6) \quad F(7, -2, 7) \quad G(10, 2, 9) \quad H(13, 3, -4)$$

- (a) Determine the exact area of P , giving your answer as a simplified surd. (3)

The line l passes through E and is perpendicular to the plane containing P

- (b) Determine an equation for l giving your answer in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$ where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors. (2)

Given that

- P is one face of a parallelepiped
- the opposite face of the parallelepiped lies in the plane Π
- the point $(25, -4, 13)$ lies in Π

- (c) determine the volume of the parallelepiped. (5)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 9 is 10 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Wednesday 19 June 2024

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/3A**

Further Mathematics

Advanced

PAPER 3A: Further Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P75686RA

©2024 Pearson Education Ltd.
F:1/1/1/1/1/1/




Pearson

1. (a) Given that

$$y = \ln(3 + x^2)$$

complete the table with the value of y corresponding to $x = 3$, giving your answer to 4 significant figures.

x	2	2.5	3	3.5	4	4.5	5
y	1.946	2.225		2.725	2.944	3.146	3.332

(1)

In part (b) you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (b) Use Simpson's rule with all the values of y in the completed table to estimate, to 3 significant figures, the value of

$$\int_2^5 \ln(3 + x^2) dx$$

(3)

- (c) Using your answer to part (b) and making your method clear, estimate the value of

$$\int_2^5 \ln \sqrt{3 + x^2} dx$$

(1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. Use algebra to determine the values of x for which

$$|x^2 - 2x| \leq x$$

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 4 marks)



3. Use L'Hospital's rule to show that

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = 0$$

(6)



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4. $\left[\begin{array}{l} \text{The Taylor series expansion of } f(x) \text{ about } x = a \text{ is given by} \\ f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^r}{r!}f^{(r)}(a) + \dots \end{array} \right]$

The curve with equation $y = f(x)$ satisfies the differential equation

$$\cos x \frac{d^2 y}{dx^2} + y^2 \frac{dy}{dx} + \sin x = 0$$

Given that $\left(\frac{\pi}{4}, 1\right)$ is a stationary point of the curve,

- (a) determine the nature of this stationary point, giving a reason for your answer. (2)
- (b) Show that $\frac{d^3 y}{dx^3} = \sqrt{2} - 2$ at this stationary point. (4)
- (c) Hence determine a series solution for y , in ascending powers of $\left(x - \frac{\pi}{4}\right)$ up to and including the term in $\left(x - \frac{\pi}{4}\right)^3$, giving each coefficient in simplest form. (2)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 8 marks)



$$y = e^{3x} \sin x$$

(a) Use Leibnitz's theorem to show that

$$\frac{d^4 y}{dx^4} = 28e^{3x} \sin x + 96e^{3x} \cos x \quad (6)$$

(b) Hence express $\frac{d^4y}{dx^4}$ in the form

$$Re^{3x} \sin(x + \alpha)$$

where R and α are constants to be determined, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 9 marks)



6. The ellipse E has equation

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

The hyperbola H has equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where a and b are positive constants.

Given that

- the eccentricity of H is the reciprocal of the eccentricity of E
- the coordinates of the foci of H are the same as the coordinates of the foci of E

determine

(i) the value of a

(ii) the value of b

(6)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

- (a) Use the substitution $t = \tan\left(\frac{\theta}{2}\right)$ to show that

$$\int \frac{1}{2 \sin \theta + \cos \theta + 2} d\theta = \int \frac{a}{(t+b)^2 + c} dt$$

where a , b and c are constants to be determined.

(3)

- (b) Hence show that

$$\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{1}{2 \sin \theta + \cos \theta + 2} d\theta = \ln\left(\frac{2\sqrt{3}}{3}\right)$$

(4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 7 is 7 marks)



8. The parabola P has equation $y^2 = 4ax$, where a is a positive constant.

The point $A(at^2, 2at)$, where $t \neq 0$, lies on P .

- (a) Use calculus to show that an equation of the tangent to P at A is

$$yt = x + at^2 \quad (3)$$

The point $B(2k^2, 4k)$ and the point $C(2k^2, -4k)$, where k is a constant, lie on P .

The tangent to P at B and the tangent to P at C intersect at the point D .

Given that the area of the triangle BCD is 432

- (b) determine the coordinates of B and the coordinates of C . (5)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 8 marks)



9. (i) The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$

The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 13 \\ 5 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}$

where λ and μ are scalar parameters.

The lines l_1 and l_2 intersect at the point P .

- (a) Determine the coordinates of P . (2)

Given that the plane Π contains both l_1 and l_2

- (b) determine a Cartesian equation for Π . (4)

- (ii) Determine a Cartesian equation for each of the two lines that

- pass through $(0, 0, 0)$
 - make an angle of 60° with the x -axis
 - make an angle of 45° with the y -axis
- (4)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 10 marks)



10. The motion of a particle P along the x -axis is modelled by the differential equation

$$t^2 \frac{d^2 x}{dt^2} - 2t(t+1) \frac{dx}{dt} + 2(t+1)x = 8t^3 e^t \quad (\text{I})$$

where P has displacement x metres from the origin O at time t minutes, $t > 0$

- (a) Show that the transformation $x = tu$ transforms the differential equation (I) into the differential equation

$$\frac{d^2u}{dt^2} - 2\frac{du}{dt} = 8e^t \quad (4)$$

Given that P is at O when $t = \ln 3$ and when $t = \ln 5$

- (b) determine the particular solution of the differential equation (I)



DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 10 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Thursday 22 June 2023

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/3A**

Further Mathematics

Advanced

PAPER 3A: Further Pure Mathematics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P72796RA

©2023 Pearson Education Ltd.
N:1/1/1/1/




Pearson

1. (a) Use Simpson's rule with 4 intervals to find an estimate for

$$\int_0^2 e^{\sin^2 x} dx$$

Give your answer to 3 significant figures.

(4)

Given that $\int_0^2 e^{\sin^2 x} dx = 3.855$ to 4 significant figures,

- (b) comment on the accuracy of your answer to part (a).

(1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 5 marks)



2. The vertical height, h m, above horizontal ground, of a passenger on a fairground ride, t seconds after the ride starts, where $t \leq 5$, is modelled by the differential equation

$$t^2 \frac{d^2 h}{dt^2} - 2t \frac{dh}{dt} + 2h = t^3 \quad (\text{I})$$

- (a) Given that $t = e^x$, show that

$$(i) \quad t \frac{dh}{dt} = \frac{dh}{dx}$$

$$(ii) \quad t^2 \frac{d^2 h}{dt^2} = \frac{d^2 h}{dx^2} - \frac{dh}{dx} \quad (4)$$

- (b) Hence show that the transformation $t = e^x$ transforms equation (I) into the equation

$$\frac{d^2h}{dx^2} - 3\frac{dh}{dx} + 2h = e^{3x} \quad (1)$$

- (c) Hence show that

$$h = At + Bt^2 + \frac{1}{2}t^3$$

where A and B are constants.

(6)

Given that when $t = 1$, $h = 2.5$ and when $t = 2$, $\frac{dh}{dt} = -1$

- (d) determine the height of the passenger above the ground 5 seconds after the start of the ride.

(5)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 16 marks)



3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

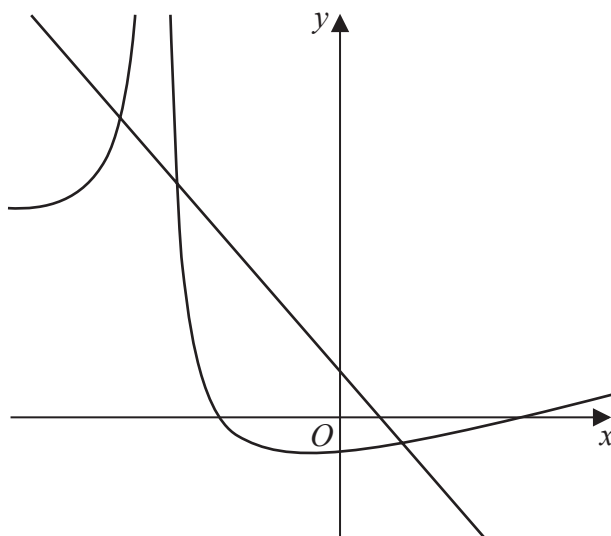


Figure 1

Figure 1 shows a sketch of the curve with equation $y = \frac{x^2 - 2x - 24}{|x + 6|}$ and the line with equation $y = 5 - 4x$

Use algebra to determine the values of x for which

$$\frac{x^2 - 2x - 24}{|x + 6|} < 5 - 4x$$

(7)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



4. The ellipse E has equation

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

- (a) Determine the exact value of the eccentricity of E (2)

The points $P(4\cos\theta, 3\sin\theta)$ and $Q(4\cos\theta, -3\sin\theta)$ lie on E where $0 < \theta < \frac{\pi}{2}$
 The line l_1 is the normal to E at the point P

- (b) Use calculus to show that l_1 has equation

$$4x \sin \theta - 3y \cos \theta = 7 \sin \theta \cos \theta \quad (4)$$

The line l_2 passes through the origin and the point Q
 The lines l_1 and l_2 intersect at the point R

- (c) Determine, in simplest form, the coordinates of R
- (4)**

- (d) Hence show that, as θ varies, R lies on an ellipse which has the same eccentricity as ellipse E (2)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 12 marks)



5. (a) Show that the substitution $t = \tan\left(\frac{x}{2}\right)$ transforms the integral

$$\int \frac{1}{2 \sin x - \cos x + 5} dx$$

into the integral

$$\int \frac{1}{3t^2 + 2t + 2} dt$$

(4)

- (b) Hence determine

$$\int \frac{1}{2 \sin x - \cos x + 5} dx$$

(4)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 12 marks)



7. With respect to a fixed origin O the point A has coordinates $(3, 6, 5)$ and the line l has equation

$$(\mathbf{r} - (12\mathbf{i} + 30\mathbf{j} + 39\mathbf{k})) \times (7\mathbf{i} + 13\mathbf{j} + 24\mathbf{k}) = \mathbf{0}$$

The points B and C lie on l such that $AB = AC = 15$

Given that A does not lie on l and that the x coordinate of B is negative,

- (a) determine the coordinates of B and the coordinates of C (4)
- (b) Hence determine a Cartesian equation of the plane containing the points A , B and C (3)

The point D has coordinates $(-2, 1, \alpha)$, where α is a constant.

Given that the volume of the tetrahedron $ABCD$ is 147

- (c) determine the possible values of α (4)
- Given that $\alpha > 0$
- (d) determine the shortest distance between the line l and the line passing through the points A and D , giving your answer to 2 significant figures. (4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

--	--	--	--	--

--	--	--	--

Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/3A

Further Mathematics

Advanced

PAPER 3A: Further Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P65497A

©2022 Pearson Education Ltd.

Q:1/1/1/1/



Pearson

1. An ellipse has equation $\frac{x^2}{16} + \frac{y^2}{4} = 1$ and eccentricity e_1

A hyperbola has equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and eccentricity e_2

Given that $e_1 \times e_2 = 1$

- (a) show that $a^2 = 3b^2$

(4)

Given also that the coordinates of the foci of the ellipse are the same as the coordinates of the foci of the hyperbola,

- (b) determine the equation of the hyperbola.

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. During 2029, the number of hours of daylight per day in London, H , is modelled by the equation

$$H = 0.3 \sin \left(\frac{x}{60} \right) - 4 \cos \left(\frac{x}{60} \right) + 11.5 \quad 0 \leq x < 365$$

where x is the number of days after 1st January 2029 and the angle is in radians.

- (a) Show that, according to the model, the number of hours of daylight in London on the 31st January 2029 will be 8.13 to 3 significant figures.

(1)

- (b) Use the substitution $t = \tan\left(\frac{x}{120}\right)$ to show that H can be written as

$$H = \frac{at^2 + bt + c}{1 + t^2}$$

where a , b and c are constants to be determined.

(2)

- (c) Hence determine, according to the model, the date of the first day of 2029 when there will be at least 12 hours of daylight in London.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 7 marks)



3. With respect to a fixed origin O , the points A and B have coordinates $(2, 2, -1)$ and $(4, 2p, 1)$ respectively, where p is a constant.

For each of the following, determine the possible values of p for which,

- (a) OB makes an angle of 45° with the positive x -axis (3)

- (b) $\vec{OA} \times \vec{OB}$ is parallel to $\begin{pmatrix} 4 \\ -p \\ 2 \end{pmatrix}$ (3)

- (c) the area of triangle OAB is $3\sqrt{2}$ (3)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 9 marks)



4. The velocity $v \text{ ms}^{-1}$, of a raindrop, t seconds after it falls from a cloud, is modelled by the differential equation

$$\frac{dv}{dt} = -0.1v^2 + 10 \quad t \geq 0$$

Initially the raindrop is at rest.

- (a) Use two iterations of the approximation formula $\left(\frac{dy}{dx}\right)_n \approx \frac{y_{n+1} - y_n}{h}$ to estimate the velocity of the raindrop 1 second after it falls from the cloud.

(5)

Given that the initial acceleration of the raindrop is found to be smaller than is suggested by the current model,

- (b) refine the model by changing the value of one constant.

(1)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 6 marks)



- (a) Use calculus to show that the equation of the tangent to H at the point $P\left(6t, \frac{6}{t}\right)$ is

(3)

The point $Q\left(12t, \frac{3}{t}\right)$ also lies on H .

- (2)

(c) Show that as t varies the locus of R is also a rectangular hyperbola.

(4)

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 9 marks)



6. The points P , Q and R have position vectors $\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$, $\begin{pmatrix} 3 \\ 1 \\ -5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$ respectively.

(a) Determine a vector equation of the plane that passes through the points P , Q and R , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$, where λ and μ are scalar parameters.

(2)

(b) Determine the coordinates of the point of intersection of the plane with the x -axis.

(4)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 6 marks)



7.

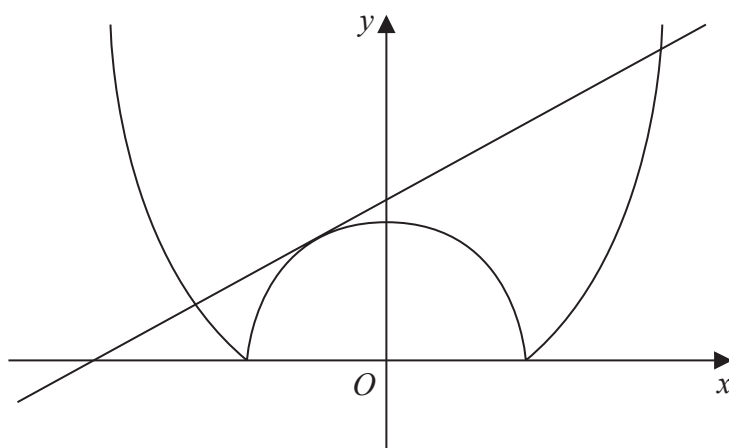


Figure 1

Figure 1 shows a sketch of the curve with equation $y = |x^2 - 8|$ and a sketch of the straight line with equation $y = mx + c$, where m and c are positive constants.

The equation

$$|x^2 - 8| = mx + c$$

has exactly 3 roots, as shown in Figure 1.

(a) Show that

$$m^2 - 4c + 32 = 0 \quad (2)$$

Given that $c = 3m$

(b) determine the value of m and the value of c (3)

(c) Hence solve

$$|x^2 - 8| \geq mx + c \quad (3)$$

DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 8 marks)



8.

$$\left[\begin{array}{l} \text{The Taylor series expansion of } f(x) \text{ about } x = a \text{ is given by} \\ f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \dots + \frac{(x - a)^r}{r!}f^{(r)}(a) + \dots \end{array} \right]$$

- (i) (a) Use differentiation to determine the Taylor series expansion of $\ln x$, in ascending powers of $(x - 1)$, up to and including the term in $(x - 1)^2$ (4)
- (b) Hence prove that

$$\lim_{x \rightarrow 1} \left(\frac{\ln x}{x - 1} \right) = 1 \quad (2)$$

- (ii) Use L'Hospital's rule to determine

$$\lim_{x \rightarrow 0} \left(\frac{1}{(x + 3) \tan(6x) \operatorname{cosec}(2x)} \right)$$

(Solutions relying entirely on calculator technology are not acceptable.) (4)



DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.



P 6 5 4 9 7 A 0 2 5 3 2

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 10 marks)



9. A particle P moves along a straight line.

At time t minutes, the displacement, x metres, of P from a fixed point O on the line is modelled by the differential equation

$$t^2 \frac{d^2 x}{dt^2} - 2t \frac{dx}{dt} + 2x + 16t^2 x = 4t^3 \sin 2t \quad (\text{I})$$

- (a) Show that the transformation $x = ty$ transforms equation (I) into the equation

$$\frac{d^2 y}{dt^2} + 16y = 4 \sin 2t \quad (5)$$

- (b) Hence find a general solution for the displacement of P from O at time t minutes. (8)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 9 continued

Handwriting practice area with horizontal lines.



Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 9 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/3A	
Further Mathematics Advanced PAPER 3A: Further Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

P66798RA

©2021 Pearson Education Ltd.

1/1/1/1/



1. The ellipse E has equation

$$\frac{x^2}{36} + \frac{y^2}{20} = 1$$

Find

- (a) the coordinates of the foci of E , (3)
- (b) the equations of the directrices of E . (2)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 5 marks)



2. (i) Use the substitution $t = \tan \frac{x}{2}$ to prove the identity

$$\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1} \equiv \sec x + \tan x \quad x \neq \frac{n\pi}{2} \quad n \in \mathbb{Z} \quad (5)$$

- (ii) Use the substitution $t = \tan \frac{\theta}{2}$ to determine the exact value of

$$\int_0^{\frac{\pi}{2}} \frac{5}{4 + 2 \cos \theta} d\theta$$

giving your answer in simplest form.

(5)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 10 marks)



3.

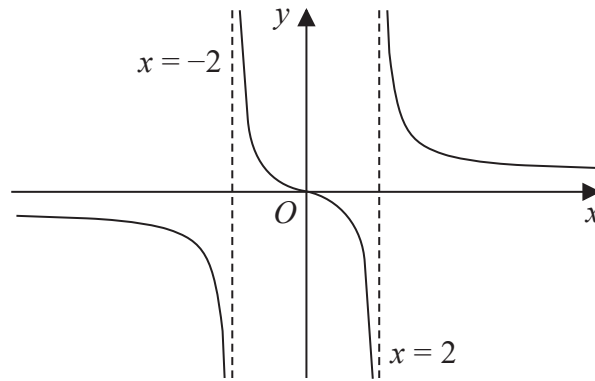


Figure 1

Figure 1 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \frac{x}{|x| - 2}$$

Use algebra to determine the values of x for which

$$2x - 5 > \frac{x}{|x| - 2}$$

(8)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

(Total for Question 3 is 8 marks)



4.

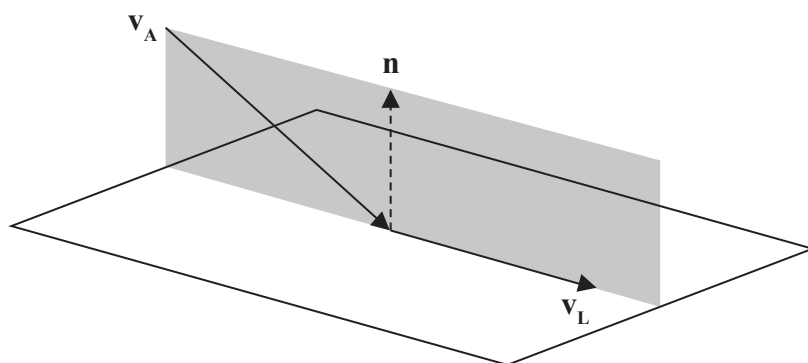


Figure 2

A small aircraft is landing in a field.

In a model for the landing the aircraft travels in different straight lines before and after it lands, as shown in Figure 2.

The vector $\mathbf{v}_A$ is in the direction of travel of the aircraft as it approaches the field.

The vector $\mathbf{v}_L$ is in the direction of travel of the aircraft after it lands.

With respect to a fixed origin, the field is modelled as the plane with equation

$$x - 2y + 25z = 0$$

and

$$\mathbf{v}_A = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$$

(a) Write down a vector $\mathbf{n}$ that is a normal vector to the field.

(1)

(b) Show that $\mathbf{n} \times \mathbf{v}_A = \lambda \begin{pmatrix} 13 \\ 19 \\ 1 \end{pmatrix}$, where λ is a constant to be determined.

(2)

When the aircraft lands it remains in contact with the field and travels in the direction $\mathbf{v}_L$.

The vector $\mathbf{v}_L$ is in the same plane as both $\mathbf{v}_A$ and $\mathbf{n}$ as shown in Figure 2.

(c) Determine a vector which has the same direction as $\mathbf{v}_L$.

(3)

(d) State a limitation of the model.

(1)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5. The parabola C has equation

$$y^2 = 32x$$

and the hyperbola H has equation

$$\frac{x^2}{36} - \frac{y^2}{9} = 1$$

- (a) Write down the equations of the asymptotes of H .

(1)

The line l_1 is normal to C and parallel to the asymptote of H with positive gradient.

The line l_2 is normal to C and parallel to the asymptote of H with negative gradient.

- (b) Determine

- (i) an equation for l_1

- (ii) an equation for l ,

(4)

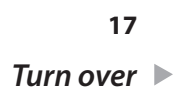
The lines l_1 and l_2 meet H at the points P and Q respectively.

- (c) Find the area of the triangle OPQ , where O is the origin.

(4)



DO NOT WRITE IN THIS AREA



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 9 marks)



6.

$$\left[\begin{array}{l} \text{The Taylor series expansion of } f(x) \text{ about } x = a \text{ is given by} \\ f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^r}{r!}f^{(r)}(a) + \dots \end{array} \right]$$

Given that

$$y = (1 + \ln x)^2 \quad x > 0$$

(a) show that $\frac{d^2 y}{dx^2} = -\frac{2 \ln x}{x^2}$ (4)

(b) Hence find $\frac{d^3y}{dx^3}$ (2)

(c) Determine the Taylor series expansion about $x = 1$ of

$$(1 + \ln x)^2$$

in ascending powers of $(x - 1)$, up to and including the term in $(x - 1)^3$

Give each coefficient in simplest form.

(3)

(d) Use this series expansion to evaluate

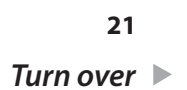
$$\lim_{x \rightarrow 1} \frac{2x - 1 - (1 + \ln x)^2}{(x - 1)^3}$$

explaining your reasoning clearly.

(3)

[illegible]

DO NOT WRITE IN THIS AREA



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7. With respect to a fixed origin O , the line l has equation

$$(\mathbf{r} - (12\mathbf{i} + 16\mathbf{j} - 8\mathbf{k})) \times (9\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}) = \mathbf{0}$$

The point A lies on l such that the direction cosines of $\overrightarrow{OA}$ with respect to the $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ axes are $\frac{3}{7}$, β and γ .

Determine the coordinates of the point A .

(7)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 7 marks)



8. A community is concerned about the rising level of pollutant in its local pond and applies a chemical treatment to stop the increase of pollutant.

The concentration, x parts per million (ppm), of the pollutant in the pond water t days after the chemical treatment was applied, is modelled by the differential equation

$$\frac{dx}{dt} = \frac{3 + \cosh t}{3x^2 \cosh t} - \frac{1}{3}x \tanh t \quad (\text{I})$$

When the chemical treatment was applied the concentration of pollutant was 3 ppm.

- (a) Use the iteration formula

$$\left(\frac{dy}{dx}\right)_n \approx \frac{(y_{n+1} - y_n)}{h}$$

once to estimate the concentration of the pollutant in the pond water 6 hours after the chemical treatment was applied.

(4)

- (b) Show that the transformation $u = x^3$ transforms the differential equation (I) into the differential equation

$$\frac{du}{dt} + u \tanh t = 1 + \frac{3}{\cosh t} \quad (\text{II})$$

(3)

- (c) Determine the general solution of equation (II)

(4)

- (d) Hence find an equation for the concentration of pollutant in the pond water t days after the chemical treatment was applied.

(3)

- (e) Find the percentage error of the estimate found in part (a) compared to the value predicted by the model, stating if it is an overestimate or an underestimate.

(3)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 17 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Thursday 15 October 2020			
Afternoon (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3A	
Further Mathematics Advanced Paper 3A: Further Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P62673A

©2020 Pearson Education Ltd.

1/1/1/1/1/1/



1. Use l'Hospital's Rule to show that

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{(e^{\sin x} - \cos(3x) - e)}{\tan(2x)} = -\frac{3}{2}$$

(5)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2.

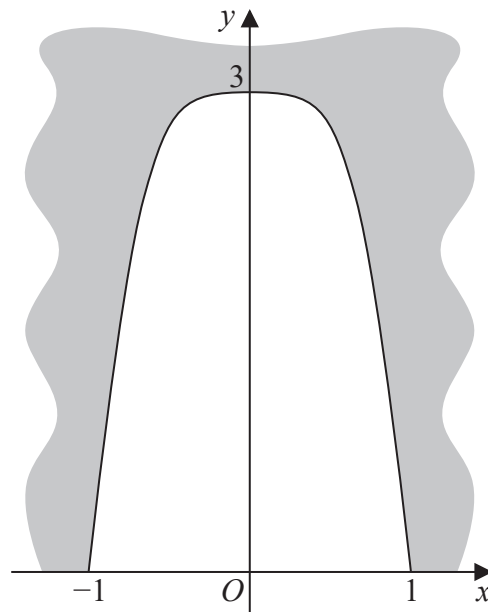


Figure 1

Figure 1 shows a sketch of the vertical cross-section of the entrance to a tunnel. The width at the base of the tunnel entrance is 2 metres and its maximum height is 3 metres.

The shape of the cross-section can be modelled by the curve with equation $y = f(x)$ where

$$f(x) = 3 \cos\left(\frac{\pi}{2}x^2\right) \quad x \in [-1, 1]$$

A wooden door of uniform thickness 85 mm is to be made to seal the tunnel entrance.

Use Simpson's rule with 6 intervals to estimate the volume of wood required for this door, giving your answer in m^3 to 4 significant figures.

(6)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3. The points A , B and C , with position vectors $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = \mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$ and $\mathbf{c} = -2\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$ respectively, lie on the plane Π

(a) Find $\vec{AB} \times \vec{AC}$ (3)

(b) Find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = p$ (2)

The point D has position vector $8\mathbf{i} + 7\mathbf{j} + 5\mathbf{k}$

(c) Determine the volume of the tetrahedron $ABCD$ (4)



DO NOT WRITE IN THIS AREA

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4.

$$f(x) = x^4 \sin(2x)$$

Use Leibnitz's theorem to show that the coefficient of $(x - \pi)^8$ in the Taylor series expansion of $f(x)$ about π is

$$\frac{a\pi + b\pi^3}{315}$$

where a and b are integers to be determined.

(8)

$$\left[\begin{array}{l} \text{The Taylor series expansion of } f(x) \text{ about } x = k \text{ is given by} \\ f(x) = f(k) + (x - k)f'(k) + \frac{(x - k)^2}{2!}f''(k) + \dots + \frac{(x - k)^r}{r!}f^{(r)}(k) + \dots \end{array} \right]$$



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 8 marks)



5. The ellipse E has equation

$$\frac{x^2}{36} + \frac{y^2}{16} = 1$$

The points S and S' are the foci of E .

- (a) Find the coordinates of S and S' (3)

- (b) Show that for any point P on E , the triangle PSS' has constant perimeter and determine its value. (4)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 7 marks)



6. A physics student is studying the movement of particles in an electric field. In one experiment, the distances in micrometres of two moving particles, A and B , from a fixed point O are modelled by

$$d_A = |5t - 31|$$

$$d_B = |3t^2 - 25t + 8|$$

respectively, where t is the time in seconds after motion begins.

- (a) Use algebra to find the range of time for which particle A is further away from O than particle B is from O .

(8)

It was recorded that the distance of particle B from O was less than the distance of particle A from O for approximately 4 seconds.

- (b) Use this information to assess the validity of the model.

(2)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



7. The points $P(9p^2, 18p)$ and $Q(9q^2, 18q)$, $p \neq q$, lie on the parabola C with equation

$$y^2 = 36x$$

The line l passes through the points P and Q

- (a) Show that an equation for the line l is

$$(p + q)y = 2(x + 9pq) \tag{3}$$

The normal to C at P and the normal to C at Q meet at the point A .

- (b) Show that the coordinates of A are

$$(9(p^2 + q^2 + pq + 2), -9pq(p + q)) \quad (7)$$

Given that the points P and Q vary such that l always passes through the point $(12, 0)$

- (c) find, in the form $y^2 = f(x)$, an equation for the locus of A , giving $f(x)$ in simplest form.



DO NOT WRITE IN THIS AREA

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 14 marks)



8.
$$f(x) = \frac{3}{13 + 6\sin x - 5\cos x}$$

Using the substitution $t = \tan\left(\frac{x}{2}\right)$

(a) show that $f(x)$ can be written in the form

$$\frac{3(1+t^2)}{2(3t+1)^2+6} \quad (3)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$f(x) = \frac{3}{7}$$

giving your answers to 2 decimal places where appropriate. (5)

(c) Use the result of part (a) to show that

$$\int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} f(x) dx = K \left(\arctan\left(\frac{\sqrt{3}-9}{3}\right) - \arctan\left(\frac{\sqrt{3}+3}{3}\right) + \pi \right)$$

where K is a constant to be determined. (8)



DO NOT WRITE IN THIS AREA



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 16 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Thursday 13 June 2019			
Afternoon (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3A	
Further Mathematics			
Advanced			
Paper 3A: Further Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P61179A

©2019 Pearson Education Ltd.

1/1/1/1/



Answer ALL questions. Write your answers in the spaces provided.

1. Use Simpson's rule with 4 intervals to estimate

$$\int_{0.4}^2 e^{x^2} dx$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. Given that k is a real non-zero constant and that

$$y = x^3 \sin kx$$

use Leibnitz's theorem to show that

$$\frac{d^5 y}{dx^5} = (k^2 x^2 + A)k^3 x \cos kx + B(k^2 x^2 + C)k^2 \sin kx$$

where A , B and C are integers to be determined.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3.

$$\frac{dy}{dx} = x - y^2 \quad (\text{I})$$

(a) Show that

$$\frac{d^5 y}{dx^5} = ay \frac{d^4 y}{dx^4} + b \frac{dy}{dx} \frac{d^3 y}{dx^3} + c \left(\frac{d^2 y}{dx^2} \right)^2$$

where a , b and c are integers to be determined.

(4)

(b) Hence find a series solution, in ascending powers of x as far as the term in x^5 , of the differential equation (I), given that $y = 1$ at $x = 0$

(5)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



4. The parabola C has equation

$$y^2 = 16x$$

The distinct points $P(p^2, 4p)$ and $Q(q^2, 4q)$ lie on C , where $p \neq 0$, $q \neq 0$

The tangent to C at P and the tangent to C at Q meet at the point $R(-28, 6)$.

Show that the area of triangle PQR is 1331

(8)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 8 marks)



5.

$$I = \int \frac{1}{4\cos x - 3\sin x} dx \quad 0 < x < \frac{\pi}{4}$$

Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that

$$I = \frac{1}{5} \ln \left(\frac{2 + \tan\left(\frac{x}{2}\right)}{1 - 2\tan\left(\frac{x}{2}\right)} \right) + k$$

where k is an arbitrary constant.

(8)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6. The concentration of a drug in the bloodstream of a patient, t hours after the drug has been administered, where $t \leq 6$, is modelled by the differential equation

$$t^2 \frac{d^2 C}{dt^2} - 5t \frac{dC}{dt} + 8C = t^3 \quad (\text{I})$$

where C is measured in micrograms per litre.

- (a) Show that the transformation $t = e^x$ transforms equation (I) into the equation

$$\frac{d^2C}{dx^2} - 6\frac{dC}{dx} + 8C = e^{3x} \quad (\text{II})$$

- (b) Hence find the general solution for the concentration C at time t hours.

Given that when $t = 6$, $C = 0$ and $\frac{dC}{dt} = -36$

- (c) find the maximum concentration of the drug in the bloodstream of the patient.



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

[illegible]

7. With respect to a fixed origin O , the points A , B and C have coordinates $(3, 4, 5)$, $(10, -1, 5)$ and $(4, 7, -9)$ respectively.

The plane Π has equation $4x - 8y + z = 2$

The line segment AB meets the plane Π at the point P and the line segment BC meets the plane Π at the point Q .

- (a) Show that, to 3 significant figures, the area of quadrilateral $APQC$ is 38.5 (6)

The point D has coordinates $(k, 4, -1)$, where k is a constant.

Given that the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ form three edges of a parallelepiped of volume 226

- (b) find the possible values of the constant k . (4)



DO NOT WRITE IN THIS AREA



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. The hyperbola H has equation

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

The line l_1 is the tangent to H at the point $P(4\cosh \theta, 3\sinh \theta)$.

The line l_1 meets the x -axis at the point A .

The line l_2 is the tangent to H at the point $(4, 0)$.

The lines l_1 and l_2 meet at the point B and the midpoint of AB is the point M .

- (a) Show that, as θ varies, a Cartesian equation for the locus of M is

$$y^2 = \frac{9(4-x)}{4x} \quad p < x < q$$

where p and q are values to be determined.

(11)

Let S be the focus of H that lies on the positive x -axis.

- (b) Show that the distance from M to S is greater than 1

(3)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
Time: 1 hour 30 minutes		Paper Reference 9FM0/3A	
Further Mathematics Advanced Paper 3A: Further Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S61306A

©2018 Pearson Education Ltd.

1/1/1/



Answer ALL questions. Write your answers in the spaces provided.

1. Use algebra to find the range of values of x for which

$$2x - 5 > \frac{3}{x}$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



- 3.** The line with equation $y = mx + c$ is a tangent to the parabola with equation $y^2 = 4ax$

(a) Show that $a = mc$

(3)

A parabola has equation $y^2 = 4x$

(b) Find equations for the tangents to this parabola that pass through the point $(-2, 1)$

(5)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



4. The plane Π has equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix} + s \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

The point A has coordinates $(4, -3, c)$

- (a) Find, in terms of c , an equation for the line that passes through the point A and is perpendicular to the plane Π . Give your answer in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$

(3)

The point A' is a reflection of the point A in the plane Π

- (b) Find, in terms of c , the coordinates of the point A'

(5)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 8 marks)



5. A student attempts the following question.

Given that

$$\frac{d^2y}{dx^2} = x^2 e^x + y^2$$

find an expression for $\frac{d^3y}{dx^3}$

- (a) The student writes

$$\frac{d^3y}{dx^3} = 2xe^x + 2y$$

Identify all the errors in the student's response.

(2)

- (b) Given that

$$\frac{d^2y}{dx^2} = x^2 e^x + y^2$$

and that $y = 1$ and $\frac{dy}{dx} = e$ at $x = 1$, find a series expansion for y , in ascending powers of $(x - 1)$, up to and including the term in $(x - 1)^3$

(5)

- (c) Explain why this series expansion does not give a valid estimate for the value of y when $x = e$

(1)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6.

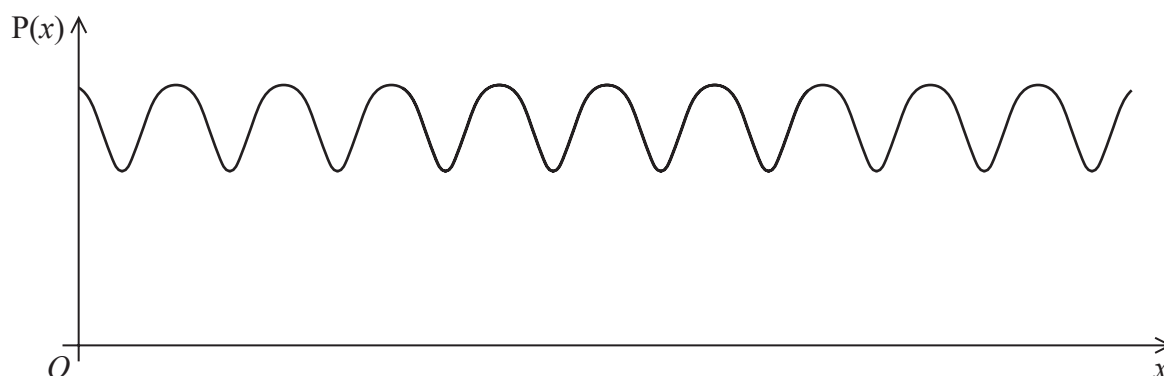


Figure 1

Figure 1 shows part of the graph of the function $P(x)$ with equation

$$P(x) = 105 - 20 \sin(6x) + 4 \cos(12x) \quad 0 < x < 10$$

(a) Using $t = \tan(3x)$, show that

$$\frac{dP}{dx} = \frac{24(t^2 - 1)(5t^2 + 8t + 5)}{(1 + t^2)^2} \quad (6)$$

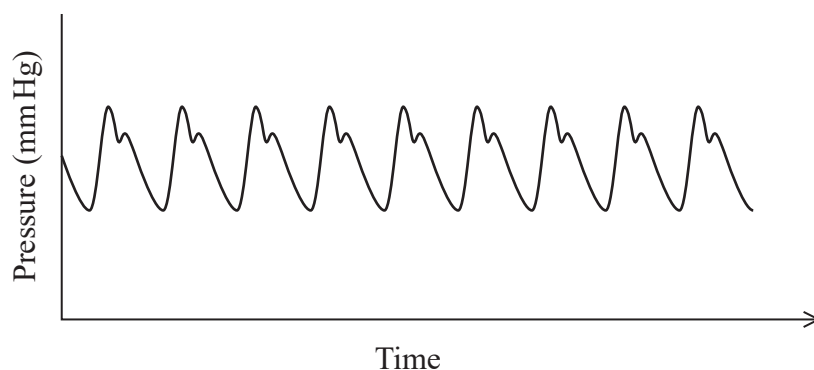


Figure 2

Figure 2 shows part of the graph of a patient's blood pressure, in millimetres of mercury (mmHg), at time x seconds.

The blood pressure of a healthy person has a minimum value between 80 and 90 mmHg, when the heart is at rest, and a maximum value between 120 and 140 mmHg when the heart beats.

The function $P(x)$ is used to model the blood pressure of the patient.

(b) Use the model and an algebraic method to determine whether or not the patient's blood pressure would be considered healthy.

(5)

(c) Use the model to determine the patient's heart rate in beats per minute.

(2)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



S 6 1 3 0 6 A 0 1 7 2 8

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



7.

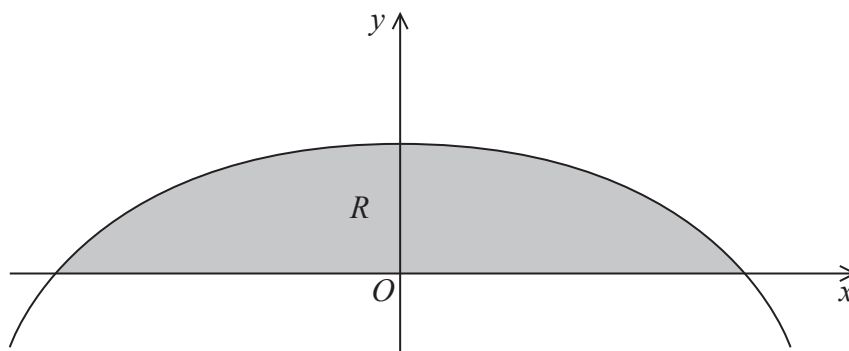


Figure 3

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{\cos x}{1 + \cos x}$$

The region R , shown shaded in Figure 1, is bounded by the curve and the x -axis.

Use calculus and the substitution $t = \tan\left(\frac{x}{2}\right)$ to show that the exact area of region R

is $a\pi + b$, where a and b are constants to be determined.

(9)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



8. At time $t = 0$, a dose of an antibiotic is given to a patient. A doctor is monitoring the concentration of the antibiotic in the patient's blood. The concentration, x , measured in mg per litre of the antibiotic in the patient's blood after t hours is modelled by the differential equation

$$2x \frac{d^2x}{dt^2} - 6 \left(\frac{dx}{dt} \right)^2 + 3x^4 + x^2 = 0$$

- (a) Show that the substitution $y = \frac{1}{x^2}$ transforms this differential equation into

$$\frac{d^2y}{dt^2} - y = 3 \quad (5)$$

When the dose of the antibiotic was given to the patient

- the concentration of the antibiotic in their blood was 2 mg per litre
- the rate of change of the concentration of the antibiotic in their blood was 1 mg per litre per hour.

- (b) Find the time it takes for the concentration of the antibiotic in the patient's blood to be halved. Give your answer to the nearest minute.

(11)



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 16 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 2			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/3A	
Further Mathematics Advanced Paper 3A: Further Pure Mathematics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S63923A

©2020 Pearson Education Ltd.

1/1/



Answer ALL questions. Write your answers in the spaces provided.

1. Given that

$$\frac{7 - 2x}{|5x - 3| + 1} < 4$$

(a) show that

$$|5x - 3| > \frac{3 - 2x}{4} \tag{2}$$

(b) On the same diagram, sketch the graph with equation $y = |5x - 3|$ and the graph with equation $y = \frac{3 - 2x}{4}$ showing the coordinates of the points where the graphs meet the coordinate axes.

(c) Use algebra to determine the exact set of values of x for which

$$\frac{7-2x}{|5x-3|+1} < 4 \tag{4}$$

giving your answer in set notation.



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. The height, y kilometres, of a rocket above horizontal ground is modelled by the differential equation

$$2\frac{d^2y}{dt^2} + y\frac{dy}{dt} - 5 = 0$$

where t is the time in minutes after the rocket is launched.

Given that three minutes after the launch, the rocket is at a height of 5 km and is ascending vertically at a rate of 0.72 kilometres per minute,

- (a) find the vertical acceleration of the rocket three minutes after the launch.

(2)

- (b) Use each of the approximations

$$\left(\frac{d^2y}{dt^2}\right)_n \approx \frac{y_{n+1} - 2y_n + y_{n-1}}{h^2} \quad \text{and} \quad \left(\frac{dy}{dt}\right)_n \approx \frac{y_{n+1} - y_{n-1}}{2h}$$

once to estimate the height of the rocket above the ground 3 minutes and 15 seconds after the launch.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 6 marks)



- 3.** The rectangular hyperbola H has equation $xy = 9$

The line l_1 with equation $y = mx + c$ is a tangent to H at the point P .

- (a) Show that

$$c^2 + km = 0$$

where k is a constant to be determined.

(2)

The parabola C has equation $y^2 = 32x$. The directrix of C crosses the x -axis at the point Q .

The line l_1 passes through Q and the line l_2 is the normal to H at the point P .

- (b) Determine the exact gradient of l_γ

(4)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA



4. Algae are plant-like organisms which are commonly found in water.

A biologist is investigating the concentration of algae in a garden pond that has recently been filled with water. The concentration of algae in the water, A , in cells per microlitre, is modelled by the differential equation

$$\frac{dA}{dt} = \frac{A(4A + t)}{t^2} \quad 0 < t \leq 125$$

where t is the time in days after the pond was filled.

- (a) Show that the substitution $A = zt$ transforms the differential equation

$$\frac{dA}{dt} = \frac{A(4A + t)}{t^2} \quad (\text{I})$$

into the differential equation

$$t \frac{dz}{dt} = 4z^2 \quad (\text{II})$$

(3)

Exactly one day after the pond was filled, the concentration of algae was 0.05 cells per microlitre.

- (b) By solving differential equation (II), determine the concentration of algae 100 days after the pond was filled, giving your answer to 3 significant figures.

(5)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5. (a) Show that an indeterminate form results when the evaluation of

$$\lim_{x \rightarrow 5} \left(\frac{-4 + \sqrt{1 + 3x}}{x^4 - 24x^2 - 25} \right)$$

is attempted by direct substitution of $x = 5$

(1)

(b) Find the Taylor series expansion of $\sqrt{1+3x}$ in ascending powers of $(x-5)$ up to and including the term in $(x-5)^2$, giving all coefficients in their simplest form.

(5)

(c) Hence determine the exact value of

$$\lim_{x \rightarrow 5} \left(\frac{-4 + \sqrt{1 + 3x}}{x^4 - 24x^2 - 25} \right)$$

(4)

(d) Showing your method clearly, use l'Hospital's rule to verify your answer to part (c).

(3)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 13 marks)



6.

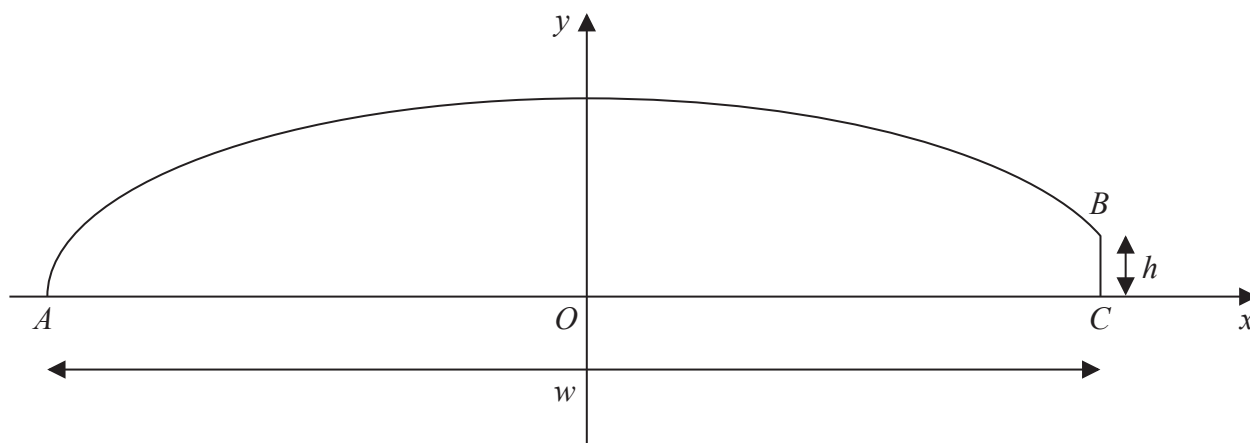


Figure 1

Figure 1 shows an architect's sketch of the cross-section of a museum.

The cross-section $ABCOA$ is modelled by the region bounded by part of the ellipse E , the vertical line BC and the x -axis. The ellipse E has equation

$$\frac{x^2}{625} + \frac{y^2}{49} = 1$$

where x and y are measured in metres.

Point C is a focus of E .

The width AC is w metres and the height BC is h metres, as shown in Figure 1.

(a) Determine the value of w and the value of h .

(5)

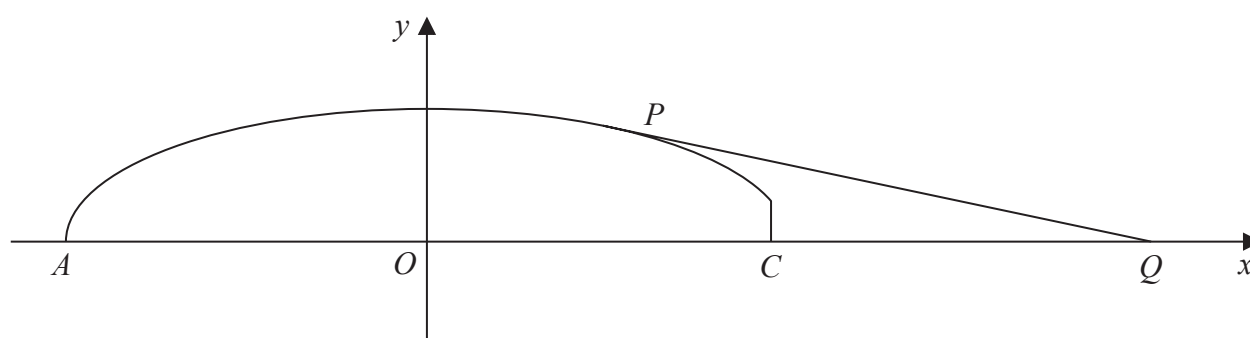


Figure 2

Figure 2 shows a sketch of the same design with a walkway, PQ , added that enables visitors to access the museum roof. PQ is the tangent to E at the point P which has coordinates $(25 \cos \theta, 7 \sin \theta)$, where $0 < \theta < \frac{\pi}{2}$

Given that $AQ = 75$ metres,

(b) show that $\theta = \frac{\pi}{3}$

(5)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



7. A line has direction ratios $a : b : c$ and direction cosines l, m and n .

(a) Show clearly that $l^2 + m^2 + n^2 = 1$

(2)

The direction vector of the line makes the same angle with all three of the positive x, y and z axes.

With respect to a fixed origin O , the point P has coordinates $(2, -1, 2)$.

Given that the line passes through P ,

(b) find an equation for the line in the form $\mathbf{r} \times \mathbf{d} = \mathbf{e}$ where $\mathbf{d}$ and $\mathbf{e}$ are vectors to be determined.

(3)

The line meets the plane with equation

$$\mathbf{r} \cdot (\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}) = 23$$

at the point Q .

(c) Determine the exact area of triangle OPQ

(6)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



S 6 3 9 2 3 A 0 2 5 3 2

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 11 marks)



8. [In this question you may assume the t -formulae for $\cos x$ and $\frac{dx}{dt}$ without proof.]

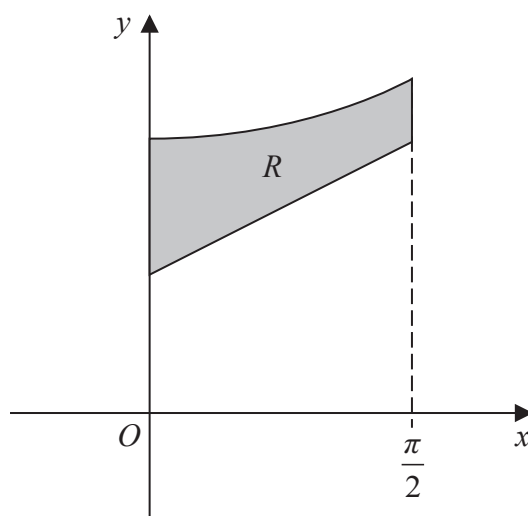


Figure 3

An engineer is designing a part for a motorcycle.

The part is modelled by the solid formed by rotating the region R , shown shaded in Figure 3, through 2π radians about the x -axis.

Region R is bounded by the y -axis, the line with equation $\pi y = 2x + \pi$, the line with equation $x = \frac{\pi}{2}$ and the curve with equation

$$y = \sqrt{\frac{12}{2 + \cos x}}$$

where x and y are measured in centimetres.

The part is to be made from steel of density 8 g cm^{-3}

- (a) Using the substitution $t = \tan\left(\frac{x}{2}\right)$ and algebraic integration, find, according to the model, the exact mass in grams of the part, giving your answer in the form

$$\frac{4}{3}\pi^2(a\sqrt{3} + b)$$

where a and b are integers to be determined.

(11)

- (b) Give a reason why the mass of steel predicted by the model may not be an accurate value for the mass of steel that would be used to make the part.

(1)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DO NOT WRITE IN THIS AREA

Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Further Mathematics

Advanced

Further Mathematics Option 1

Paper 3: Further Pure Mathematics 1

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/3A

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S54440A

©2017 Pearson Education Ltd.

1/1/1/1/



S 5 4 4 4 0 A 0 1 1 8



Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. Use Simpson's Rule with 6 intervals to estimate

$$\int_1^4 \sqrt{1+x^3} \, dx$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

2. Given k is a constant and that

$$y = x^3 e^{kx}$$

use Leibnitz theorem to show that

$$\frac{d^n y}{dx^n} = k^{n-3} e^{kx} (k^3 x^3 + 3nk^2 x^2 + 3n(n-1)kx + n(n-1)(n-2)) \quad (4)$$

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

3. A vibrating spring, fixed at one end, has an external force acting on it such that the centre of the spring moves in a straight line. At time t seconds, $t \geq 0$, the displacement of the centre C of the spring from a fixed point O is x micrometres.

The displacement of C from O is modelled by the differential equation

$$t^2 \frac{d^2 x}{dt^2} - 2t \frac{dx}{dt} + (2 + t^2)x = t^4 \quad (\text{I})$$

- (a) Show that the transformation $x = tv$ transforms equation (I) into the equation

$$\frac{d^2 v}{dt^2} + v = t \quad (\text{II})$$

- (b) Hence find the general equation for the displacement of C from O at time t seconds.

(7)

- (c) (i) State what happens to the displacement of C from O as t becomes large.

- (ii) Comment on the model with reference to this long term behaviour.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 14 marks)

$$\frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + y = 0 \quad (\text{I})$$
$$\frac{d^5 y}{dx^5} = ax \frac{d^4 y}{dx^4} + b \frac{d^3 y}{dx^3}$$

(4)

of the differential equation (I) where $y = 0$ and $\frac{dy}{dx} = 1$ at $x = 0$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

5. The normal to the parabola $y^2 = 4ax$ at the point $P(ap^2, 2ap)$ passes through the parabola again at the point $Q(aq^2, 2aq)$.

The line OP is perpendicular to the line OQ , where O is the origin.

Prove that $p^2 = 2$

(9)

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

6. A tetrahedron has vertices $A(1, 2, 1)$, $B(0, 1, 0)$, $C(2, 1, 3)$ and $D(10, 5, 5)$.

Find

- (a) a Cartesian equation of the plane ABC .

(3)

- (b) the volume of the tetrahedron $ABCD$.

(3)

The plane Π has equation $2x - 3y + 3 = 0$

The point E lies on the line AC and the point F lies on the line AD .

Given that Π contains the point B , the point E and the point F ,

- (c) find the value of k such that $\vec{AE} = k\vec{AC}$.

(3)

Given that $\vec{AF} = \frac{1}{9}\vec{AD}$

- (d) show that the volume of the tetrahedron $ABCD$ is 45 times the volume of the tetrahedron $ABEF$.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

(Total for Question 6 is 11 marks)

7. P and Q are two distinct points on the ellipse described by the equation $x^2 + 4y^2 = 4$

The line l passes through the point P and the point Q .

The tangent to the ellipse at P and the tangent to the ellipse at Q intersect at the point (r, s) .

Show that an equation of the line l is

$$4sy + rx = 4$$

(8)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

8.

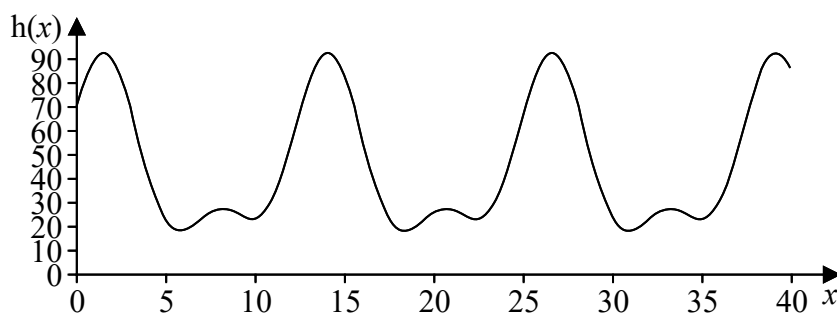


Figure 1

Figure 1 shows the graph of the function $h(x)$ with equation

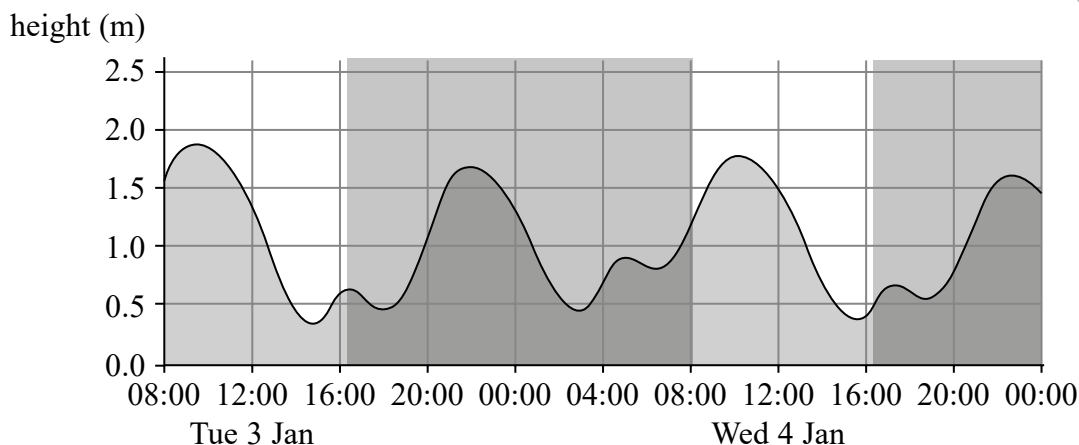
$$h(x) = 45 + 15 \sin x + 21 \sin\left(\frac{x}{2}\right) + 25 \cos\left(\frac{x}{2}\right) \quad x \in [0, 40]$$

(a) Show that

$$\frac{dh}{dx} = \frac{(t^2 - 6t - 17)(9t^2 + 4t - 3)}{2(1 + t^2)^2}$$

where $t = \tan\left(\frac{x}{4}\right)$.

(6)



Source: ¹Data taken on 29th December 2016 from <http://www.ukho.gov.uk/easytide/EasyTide>

Figure 2

Figure 2 shows a graph of predicted tide heights, in metres, for Portland harbour from 08:00 on the 3rd January 2017 to the end of the 4th January 2017¹.

The graph of $kh(x)$, where k is a constant and x is the number of hours after 08:00 on 3rd of January, can be used to model the predicted tide heights, in metres, for this period of time.

(b) (i) Suggest a value of k that could be used for the graph of $kh(x)$ to form a suitable model.

(ii) Why may such a model be suitable to predict the times when the tide heights are at their peaks, but not to predict the heights of these peaks?

(3)

(c) Use Figure 2 and the result of part (a) to estimate, to the nearest minute, the time of the highest tide height on the 4th January 2017.

(6)

DO NOT WRITE IN THIS AREA

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

--	--	--	--	--

Candidate Number

--	--	--	--	--

Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3A**

Further Mathematics
Advanced
Paper 3A: Further Pure Mathematics 1

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

S61296A

©2018 Pearson Education Ltd.

1/1/1/



S 6 1 2 9 6 A 0 1 2 8


Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. The hyperbola H has equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Show that an equation of the tangent to H at the point $P(a \sec \theta, b \tan \theta)$ is

$$ya \tan \theta = xb \sec \theta - ab$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. Relative to a fixed origin O , the points A , B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = \mathbf{i} + \mathbf{j} \quad \mathbf{b} = 3\mathbf{i} - \mathbf{j} + \mathbf{k} \quad \mathbf{c} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$$

Find

- (a) the area of triangle OBC

(2)

- (b) the volume of the tetrahedron $OABC$

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3. (a) Using the substitution $t = \tan\left(\frac{x}{2}\right)$ show that the equation

$$4 \tan x + 3 \cot\left(\frac{x}{2}\right) \sec^2\left(\frac{x}{2}\right) = 0$$

can be written as

$$3t^4 - 8t^2 - 3 = 0 \quad (3)$$

(b) Hence solve, for $-2\pi < x \leq 2\pi$

$$4 \tan x + 3 \cot\left(\frac{x}{2}\right) \sec^2\left(\frac{x}{2}\right) = 0$$

giving your answers in terms of π

6



Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



4. Giving your answer using set notation, use algebra to find the complete set of values of x for which

$$|x^2 - 2| > 2x$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 4 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 4 is 5 marks)



5. (i) Given that $\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors, find the value of

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{a})$$

giving a reason for your answer.

(2)

- (ii) Given vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ such that $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, where $\mathbf{a} \neq \mathbf{0}$ and $\mathbf{b} \neq \mathbf{c}$, show that

$$\mathbf{b} - \mathbf{c} = \lambda \mathbf{a}$$

where λ is a scalar, giving a reason for your answer.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 5 continued

(Total for Question 5 is 4 marks)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA**

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA**

-

Question 6 continued

Handwriting practice area with 30 horizontal lines.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 6 continued

Lined area for writing the answer to Question 6.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 6 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 6 is 13 marks)



7. A savings account has a fixed annual interest rate of $r\%$ which is compounded n times over the year.

The amount of money in this savings account after one year, £ A , is modelled by the equation

$$A = P \left(1 + \frac{r}{100n} \right)^n$$

where $\pounds P$ is the amount in the account at the start of the year.

A student has £1000 in this account at the start of the year and they neither add money to, nor withdraw money from this account.

The annual interest rate is 5%

- (a) If the interest is compounded 12 times a year, use the model to show, to the nearest penny, that the amount in the savings account after one year will be £1051.16

(1)

- (b) Use L'Hospital's rule to show that $\lim_{n \rightarrow \infty} \left(1 + \frac{r}{100n}\right)^n = e^{\frac{r}{100}}$

(6)

- (c) Use the model to find the maximum possible amount that the student could have in their savings account after one year.

(3)

[illegible]

Question 7 continued

Lined area for writing the answer to Question 7.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



S 6 1 2 9 6 A 0 1 7 2 8

Question 7 continued

Lined area for writing the answer to Question 7.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. A particle is moving in a straight line. The displacement, x m, of the particle relative to a fixed origin O , after it has been moving for t seconds, is modelled by the differential equation

$$t \frac{dx}{dt} + 2t^2x = x(2t + 1)$$

(a) Show that the transformation $x = wt$ transforms this equation into the equation

$$\frac{dw}{dt} + 2wt - 2w = 0 \quad (3)$$

(b) Hence show that the general equation for the displacement of the particle is

$$x = Ate^{2t-t^2} \quad (5)$$

When the particle has been moving for 2 seconds, it has a displacement of 10m.

(c) Find the particular solution for the displacement of the particle. (1)

(d) Find the maximum displacement of the particle, giving your answer to the nearest centimetre.

(e) Comment on the displacement of the particle in the long term, as predicted by the model, giving a reason for your answer. (2)



Question 8 continued

Handwriting practice area with 20 horizontal lines.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 8 continued

Lined area for writing the answer to Question 8.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 15 marks)



9.

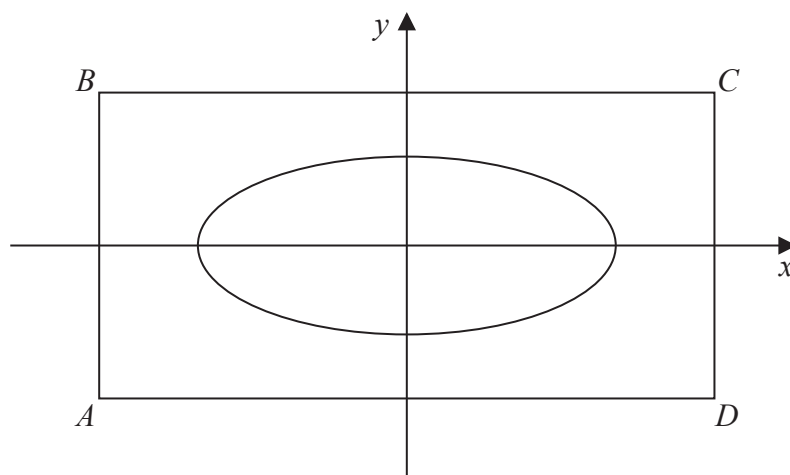


Figure 1

Figure 1 shows the plan for a rectangular garden $ABCD$.

In the middle of the garden is a large pond that may be modelled as an ellipse.

The length of the major axis of the ellipse is twice the length of the minor axis of the ellipse. The line AB and the line CD are modelled as the directrices of the ellipse.

The ellipse and the rectangle $ABCD$ lie in the same horizontal plane.

Given that the length of the garden, AD , is $\frac{16}{3}\sqrt{3}$ metres,

(a) find an equation of the ellipse.

(6)

Two water features, modelled as particles, are to be placed in the pond. The sum of the horizontal distances from the water features to any point on the edge of the pond is constant.

(b) Find the coordinates of the points at which the water features are to be placed, according to the model.

(2)

Gnomes, modelled as particles, are to be placed on the edge of the pond. Each gnome will be exactly 2 m from a water feature.

(c) Find all the possible coordinates for the gnomes.

(5)



Question 9 continued

Lined area for writing the answer to Question 9.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



S 6 1 2 9 6 A 0 2 5 2 8

Question 9 continued

Lined area for writing the answer to Question 9.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



Question 9 continued

Lined area for writing the answer to Question 9.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



S 6 1 2 9 6 A 0 2 7 2 8

Question 9 continued

Lined area for writing answers.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 9 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS

